

Ex.No.8 Use Steg-Expose to detect hidden data in images

Description

StegExpose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded. Below is a step-by-step guide on how to use StegExpose to detect hidden data in images:

Prerequisites:

1. **Java Runtime Environment (JRE):** StegExpose is a Java-based tool, so ensure you have JRE installed.
2. **StegExpose Tool:** Download StegExpose from the [official GitHub repository](#).

Step-by-Step Process:

1. Download and Set Up StegExpose:

- **Download the tool:** Go to [StegExpose GitHub](#) and download the .jar file.
- **Install Java:** Ensure that Java is installed on your machine. You can download it from the official site.
- **Prepare environment:** Place the .jar file in a folder where you plan to work on the image files.

2. Select Images for Analysis:

- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.

3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS).

```
Windows PowerShell
OpenJDK 64-Bit Server VM (Temurin)(build 25.504-b01, mixed mode)
PS C:\Users\ashra> New-Item -ItemType Directory -Path "D:\DIGITAL FORENSIC\EXP 8" -Force

Directory: D:\DIGITAL FORENSIC

Mode                LastWriteTime         Length Name
----                -
d-----          22-09-2026    08:48             EXP 8

PS C:\Users\ashra> cd "D:\DIGITAL FORENSIC\EXP 8"
PS D:\DIGITAL FORENSIC\EXP 8> pwd

Path
----
D:\DIGITAL FORENSIC\EXP 8

PS D:\DIGITAL FORENSIC\EXP 8> Get-Item "D:\DIGITAL FORENSIC\EXP 8\StegExpose.jar"

Directory: D:\DIGITAL FORENSIC\EXP 8

Mode                LastWriteTime         Length Name
----                -
-a-----          22-09-2026    08:57    3060550 StegExpose.jar
```

4. Run StegExpose on an Image:

- Use the following command to check a single image for hidden data:

bash

java -jar StegExpose.jar <image_file_path>

Example:

bash

java -jar StegExpose.jar test_image.png

- Replace <image_file_path> with the actual path of the image you want to check.

```
PS D:\DIGITAL FORENSIC\EXP 8> java -jar ".\StegExpose.jar" ".\images" default 0.2 ".\steganalysis.csv"
PS D:\DIGITAL FORENSIC\EXP 8> Get-ChildItem ".\steganalysis.csv"

Directory: D:\DIGITAL FORENSIC\EXP 8

Mode                LastWriteTime         Length Name
----                -
-a-----          22-09-2026    09:07         266 steganalysis.csv

PS D:\DIGITAL FORENSIC\EXP 8> Get-Content ".\steganalysis.csv"

File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pa
irs,RS analysis,Fusion (mean)
test.jpg.png,false,8931,0.0243849196836966,0.0019579394447364813,null,0.018467244322264695,0.014936701150232592
PS D:\DIGITAL FORENSIC\EXP 8> |
```

5. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- **Output Explanation:**
 - StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.

- **Threshold values** (suggested by the tool):
 - **Less than 0.2:** Image is clean (no hidden data).
 - **0.2 - 0.3:** Possibly some hidden data.
 - **Above 0.3:** Likely that steganography is present.

```
PS D:\DIGITAL FORENSIC\EXP 8> java -jar ".\StegExpose.jar" ".\images" default 0.2 ".\steganalysis.csv"
PS D:\DIGITAL FORENSIC\EXP 8> Get-ChildItem ".\steganalysis.csv"

Directory: D:\DIGITAL FORENSIC\EXP 8

Mode                LastWriteTime         Length Name
----                -
-a-----         22-09-2026         09:07         266 steganalysis.csv

PS D:\DIGITAL FORENSIC\EXP 8> Get-Content ".\steganalysis.csv"
File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pa
irs,RS analysis,Fusion (mean)
test.jpg.png,false,8931,0.0243849196836966,0.0019579394447364813,null,0.018467244322264695,0.014936701150232592
PS D:\DIGITAL FORENSIC\EXP 8> |
```

6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder. For example:

bash

java -jar StegExpose.jar <folder_path>

- Replace <folder_path> with the path to the folder containing the images.

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

bash

java -jar StegExpose.jar --help

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Example:

bash

java -jar StegExpose.jar suspect_image.png

Output:

makefile

Analyzing suspect_image.png...

Result: 0.4

Steganography likely present

This output indicates that hidden data is likely present in the image
suspect_image.png.